

Lab Assignment 02

New York Taxi Fare Prediction Using Deep Feedforward Neural Networks

1. Problem Statement

Develop a Deep Learning-based regression system to predict the fare amount of a New York City taxi trip using trip-related, temporal, and geolocation features.

The project should investigate the relationship between factors such as **pickup/drop-off locations, trip time, passenger count, and trip distance** and the resulting taxi fare. Students must perform EDA, data preprocessing, feature engineering, feature scaling, develop a deep feedforward neural network, evaluate its performance, and deploy the trained model.

2. Aim

To develop and deploy a Deep Feedforward Neural Network for predicting New York City taxi fares using historical taxi trip data.

3. Objectives

Students should be able to:

1. Understand the structure and characteristics of a real-world taxi fare dataset.
2. Perform systematic exploratory data analysis.
3. Visualize and interpret geolocation-based taxi trip data.
4. Analyze ridership patterns across days and hours.
5. Identify and handle missing values and data anomalies.
6. Engineer temporal and geolocation features.
7. Apply appropriate feature scaling techniques.
8. Understand deep feedforward neural networks for regression.
9. Select an appropriate loss function for a regression problem.
10. Build and train a deep neural network for fare prediction.
11. Evaluate and interpret regression model performance.
12. Deploy the trained model for predicting fares for new taxi trips.

4. Dataset

Recommended Dataset

Students may use the **NYC Taxi Fare Prediction** dataset containing historical taxi trips and fare amounts.

Dataset source: Kaggle – New York City Taxi Fare Prediction

<https://www.kaggle.com/competitions/new-york-city-taxi-fare-prediction>

Important Dataset Attributes

Typical attributes include:

- **key**
- **fare_amount**
- **pickup_datetime**
- **pickup_longitude**
- **pickup_latitude**
- **dropoff_longitude**

- `dropoff_latitude`
- `passenger_count`

Target Variable

- `fare_amount`

The target is a continuous numerical variable, making this a **regression problem**.

5. Dataset Understanding

Students should investigate the dataset before beginning modeling.

Tasks

- Load the dataset using Pandas.
- Display the first and last few records.
- Determine the number of observations and features.
- Identify data types.
- Generate descriptive statistics.
- Identify the target variable.
- Determine the range of taxi fares.
- Analyze passenger counts.
- Examine pickup and drop-off coordinates.
- Examine the date/time information.
- Identify the units and meaning of each feature.

Expected Output

Prepare a dataset summary table.

6. Exploratory Data Analysis

Perform systematic EDA to understand the characteristics and patterns in the taxi dataset.

6.1 General Dataset Analysis

Students should:

- Examine dataset dimensions.
- Examine data types.
- Generate descriptive statistics.
- Identify missing values.
- Analyze unique values.
- Examine the distribution of the target variable.
- Identify unusual observations.

6.2 Visualizations

Generate appropriate:

- Histograms
- Box plots
- Distribution plots
- Bar charts

7. Visualizing Geolocation Data

Analyze the geographical distribution of taxi trips.

7.1 Tasks

Students should:

- Plot pickup locations using latitude and longitude.
- Plot drop-off locations.
- Visualize pickup and drop-off locations together where appropriate.
- Identify regions with high concentrations of trips.
- Investigate unusual coordinates.
- Examine the relationship between geographical location and fare.

Suggested Visualizations

- Scatter plot of pickup coordinates
- Scatter plot of drop-off coordinates
- Pickup density visualization
- Drop-off density visualization
- Pickup vs drop-off location visualization

7.2 Discussion

Students should discuss:

- Which areas have the highest concentration of pickups?
- Are pickup and drop-off locations distributed uniformly?
- Are there geographical regions with unusually high fares?
- What types of coordinate values may indicate data anomalies?

8. Ridership by Day and Hour

Analyze temporal patterns in taxi usage.

8.1 Tasks

Extract information from **pickup_datetime** and investigate:

- Trips by hour
- Trips by day of week
- Trips by month
- Trips by year, if applicable
- Weekday vs weekend ridership
- Peak and off-peak periods

Required Visualizations

- Trips vs hour
- Trips vs day of week
- Trips vs month
- Heatmap of day of week $\times$ hour

8.2 Discussion

Students should identify:

- Peak travel hours
- Low-demand periods
- Differences between weekdays and weekends
- Potential relationship between demand and taxi fares

9. Data Preprocessing

Prepare the raw dataset for model development.

9.1 Missing Values

Students should:

- Identify missing values in each feature.

- Calculate the percentage of missing values.
- Determine an appropriate strategy for handling missing values.
- Justify whether records should be removed or values should be imputed.

9.2 Data Anomalies

Identify and investigate anomalous observations.

Students should examine:

- Negative fare amounts
- Zero or extremely low fares
- Extremely high fares
- Invalid passenger counts
- Invalid latitude values
- Invalid longitude values
- Pickup/drop-off locations outside the expected geographical region
- Duplicate records, where applicable

Important Requirement

- Students should not remove anomalous observations blindly.
- Each removal or correction must be supported by a reasonable justification.

10. Feature Engineering

Create meaningful features that can improve taxi fare prediction.

10.1 Temporal Features

Extract useful information from the pickup timestamp.

Students should consider:

- Hour
- Day
- Day of week
- Month
- Year
- Weekend indicator
- Rush-hour indicator

Students should investigate whether temporal features improve model performance.

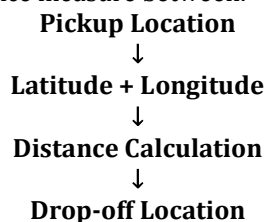
10.2 Geolocation Features

Create meaningful features from pickup and drop-off coordinates.

Required Feature

- Approximate trip distance

Students should calculate an appropriate distance measure between:



Students may investigate:

- Haversine distance
- Approximate Euclidean distance
- Latitude difference
- Longitude difference

Additional Features

Students may explore:

- Pickup/drop-off coordinate differences
- Distance per passenger
- Location-based features
- Pickup and drop-off zones

Students must explain the rationale behind each engineered feature.

12. Feature Analysis

After feature engineering:

- Examine correlations between numerical features.
- Analyze the relationship between trip distance and fare.
- Analyze the relationship between time-related variables and fare.
- Identify potentially redundant features.
- Investigate whether engineered features improve predictive performance.

13. Feature Scaling

Apply an appropriate feature scaling technique before training the neural network.

Students should investigate:

- Standardization using **StandardScaler**
- Min-Max normalization using **MinMaxScaler**

Tasks

- Explain why feature scaling is important for neural networks.
- Fit the scaler using training data only.
- Transform training, validation, and test data.
- Compare feature distributions before and after scaling.

14. Train, Validation and Test Split

Divide the dataset into appropriate subsets.

- Recommended:
 - Training Set → 70–80%
 - Validation Set → 10–15%
 - Test Set → 10–15%

Students must:

- Use a reproducible random seed.
- Prevent data leakage.
- Ensure that the test set remains unseen during model development.
- Apply preprocessing based only on the training data.

15. Deep Feedforward Neural Networks

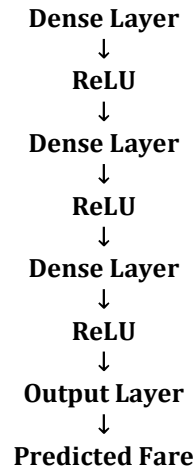
Develop a Deep Feedforward Neural Network (DNN/MLP) for taxi fare prediction.

15.1 Model Architecture

Students should design and justify a suitable architecture.

A possible starting architecture:

Input Features
↓



Students should experiment with:

- Number of hidden layers
- Number of neurons
- Activation functions
- Dropout
- Batch normalization, if appropriate
- Output layer configuration

Important

Students should **justify the final architecture** rather than simply using a predefined architecture.

16. Loss Functions for Regression

Since taxi fare is a continuous numerical value, students must investigate appropriate regression loss functions.

Loss functions to study

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- Huber Loss

Students should:

1. Explain the mathematical concept behind each loss function.
2. Discuss the effect of outliers on each loss.
3. Select an appropriate loss function for taxi fare prediction.
4. Justify the selected loss function.

17. Model Building

Develop the final neural network model.

Tasks

- Define the architecture.
- Compile the model.
- Select an optimizer.
- Select the learning rate.
- Select the loss function.
- Select appropriate evaluation metrics.
- Train the model.
- Use validation data during training.
- Record training and validation loss.

Possible Optimizers

Students may investigate:

- SGD
- Adam
- RMSprop

18. Model Training Analysis

Students should plot:

- Training loss vs epoch
- Validation loss vs epoch
- Training metric vs epoch
- Validation metric vs epoch

Analyze

- Whether the model is learning effectively.
- Whether overfitting occurs.
- Whether underfitting occurs.
- At which point validation performance stops improving.
- Whether early stopping is required.

19. Hyperparameter Tuning

Students should investigate whether tuning improves the performance of the neural network.

Hyperparameters to consider

Hyperparameter	Example Values
Hidden layers	2, 3, 4
Neurons	32, 64, 128, 256
Learning rate	0.001, 0.0005, 0.01
Batch size	16, 32, 64, 128
Dropout	0.0, 0.2, 0.3, 0.5
Optimizer	Adam, RMSprop, SGD
Loss function	MSE, MAE, Huber
Epochs	50, 100, 200

Students may use:

- Manual search
- Grid Search
- Random Search
- Bayesian Optimization
- **KerasTuner**
- **Optuna**

20. Model Evaluation

Evaluate the final model using appropriate regression metrics.

Required Metrics

- MAE
- MSE
- RMSE
- R^2 Score

Students should explain what each metric measures.

Required Visualizations

- Actual vs Predicted Fare
- Residual plot
- Prediction error distribution
- Training/validation loss curves

21. Results and Discussion

Students should present and analyse the results.

The discussion should include:

- Final model architecture
- Selected loss function
- Selected optimizer
- Hyperparameters
- Training performance
- Validation performance
- Test performance
- MAE
- RMSE
- R^2
- Actual vs predicted visualization
- Error analysis
- Effect of feature engineering
- Effect of feature scaling
- Effect of hyperparameter tuning

22. Deployment

Deploy the trained taxi fare prediction model as a simple application.

22.1 User Input

The application should allow the user to enter relevant trip information, such as:

- Pickup location
- Drop-off location
- Date
- Time
- Passenger count

The application should internally generate the required engineered features.

22.2 Prediction

The application should display:

- Estimated Taxi Fare: \$XX.XX

Optionally:

- Trip Distance: XX km
- Estimated Fare: \$XX.XX

23. Report preparation

Prepare the lab report as suggested in Lab Assignment 01.

24. Final Deliverables

- i. Jupyter Notebook / Google Colab Notebook

- ii. Dataset or dataset access link
- iii. Preprocessing pipeline
- iv. EDA visualizations
- v. Multiple model implementations
- vi. MLP implementation
- vii. Hyperparameter tuning results
- viii. Model comparison table
- ix. Final trained model
- x. Deployment application/source code
- xi. Screenshots of deployment
- xii. Lab report

Key academic requirement: Students should demonstrate the complete workflow of building a **deep learning regression application**, from understanding raw taxi data through EDA, feature engineering, neural-network modeling, evaluation, and deployment.